

Language is a skill acquired during early childhood that continues to develop for many years. Communication using language involves multiple complex processes, including listening, speaking, comprehension, and cognition. *Receptive language* ("language input") involves language comprehension, while *expressive language* ("language output") involves the ability to produce intelligible language. In normally developing children, receptive language emerges first, followed by expressive language.

To examine how age plays a role in the activation of the neurological networks involved in processing more complex language, researchers used functional magnetic resonance imaging (fMRI) to analyze how the use of receptive language generates metabolic activity in the brains of children. A total of 672 participants were divided into three groups by age: Age 4–6 ($n = 219$), age 7–9 ($n = 224$), and age 10–12 ($n = 229$).

Participants in each group were asked to listen to pre-recorded stories while undergoing fMRI of the brain to determine if an age-related difference could be found in activated brain tissue volume during the listening task. The recorded fMRI-monitored listening portion of the experiment lasted approximately 5 minutes. Immediately following this listening exercise, fMRI monitoring was discontinued and the children were administered tests to determine comprehension of the material. None of the children had been informed that they would be taking a comprehension test prior to its actual administration.

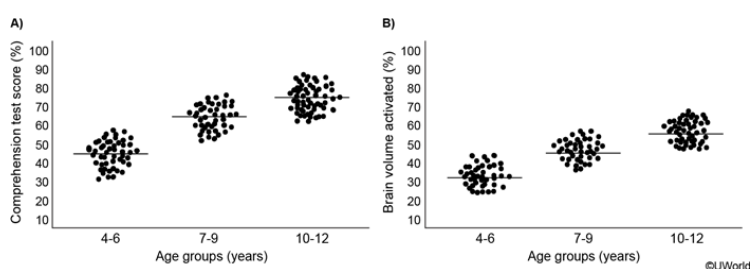


Figure 1 fMRI listening task analyzed per age group: (A) comprehension test scores, and (B) brain volume activated (Note:

Solid lines represent the mean in each group.)

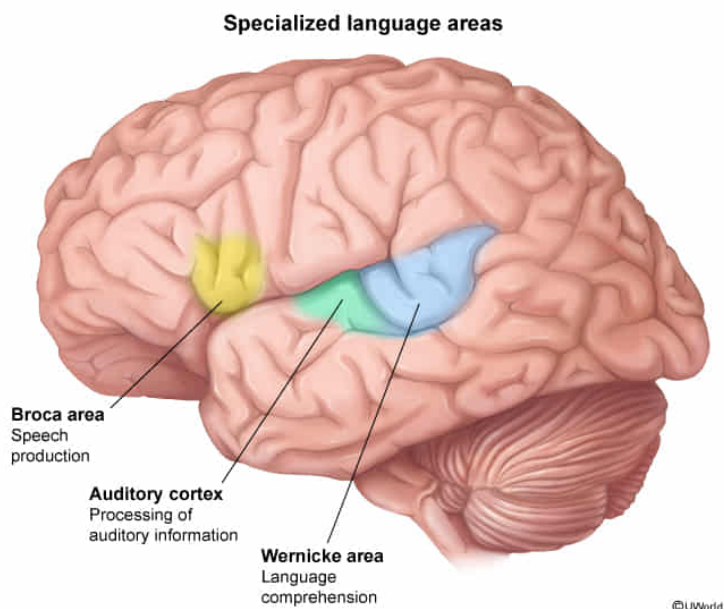
Which of the following brain regions should demonstrate the greatest degree of activation on fMRI during the task described in this study?

- ☐ A. Primary visual cortex
- ☐ B. Somatosensory cortex [3%]
- ✓ ☒ C. Wernicke area [89%]
- ☐ D. Broca area [6%]

Correct

88% answered correctly

Explanation:



According to the third paragraph, fMRI was only conducted while the children were *listening* to pre-recorded stories. **Wernicke area**, located near the primary auditory cortex (which processes auditory information) in the temporal lobe, is the brain region responsible primarily for **language comprehension** and would be activated while the children listened to the story.

(Choice A) The primary visual cortex, located in the occipital lobe, is

mainly responsible for the processing of visual input. Because the children were listening to a story rather than reading it, the primary visual cortex should not demonstrate much activation on fMRI.

(Choice B) The somatosensory cortex, located in the parietal lobe, is primarily responsible for integrating sensory input from the body about touch sensation and position of limbs in space. This part of the brain does not process hearing, so it should not demonstrate much activation on fMRI.

(Choice D) Broca area, located in the frontal lobe, is responsible primarily for **speech production**, including the motor control of the mouth that is necessary for fluent language production. During this study, fMRI was conducted only while the children were *listening* to the story, not while they were speaking. Therefore, Broca area should not demonstrate much activation on fMRI.

Educational objective:

Wernicke area is the brain region primarily responsible for language comprehension (understanding both written and auditory language input), while Broca area is the brain region primarily responsible for speech production (language output).

Time spent:0 sec
Subject:
Behavioral Sciences

QID:400392
System:
Learning, Memory, and Cognition